

- Which is better – engaging directly with CSPs, or using some cloud service aggregator or broker for providing the solutions?

Additionally, there can be queries on technical perspectives, such as:

- What are the cloud services that can be fitted to the solutions needed?
- Will a service of one CSP support similar services of another CSP?
- Which is preferred – public, private, hybrid or multi-cloud infrastructure?
- How to utilise cloud services such as PaaS, IaaS or SaaS?

There can be many such other questions while moving towards cloud based infrastructure or solutions.

Cloud enablement challenges

An enterprise could face various challenges during cloud migration, even though it is aligned for migrating to cloud services. These ground level challenges can be the parameters for defining the success factor of its cloud enablement strategies. The following are some of the challenges faced by an enterprise during cloud migration.

- **Adaption by people:** For any enterprise, technology may not be the bottleneck; the highest level of challenges can come from people who take time to adapt to it.
- **Strategies and roadmap:** Business objectives have to align with cloud strategies and define the roadmaps when moving forward. Stockholders' consent is required to overcome the psychological bottlenecks in understanding cloud services. People can be reluctant to move to cloud services due to their perceptions of security, privacy and technology.
- **Existing technical encumbrances:** These encumbrances should be identified, followed by the determination of a solution to make the move to cloud services. The legacy system should be analysed, and a plan for migration to the cloud formulated.

- **Training, maintenance and support:** Training, maintenance and support play an important part in cloud migrations. It is imperative to educate all personnel about the system and the advantages of using it. Employees have to be trained with the required skillsets. Overall maintenance and support for the system is an ongoing process.
- **ROI and payback period:** The cost associated with cloud migration should be identified and fall within the budget of an enterprise. It is important to estimate the ROI and payback period.
- **Technology stack:** Migration towards the right set of technologies plays an important role. A lot is changing around the world from the perspective of technology, and it is important to keep a measure of the right set of cloud services and solutions with a focus on the enterprise, people and process capabilities to run for the long term.
- **Security and compliance:** Security and compliance measures have to be identified and implemented with cloud solutions.
- **Downtime:** There should be a near-to-zero downtime while migrating to cloud services. It should be at acceptable measures. Downtime can have a huge cost for any enterprise.
- **Associated risk:** Identification of proper risks before migration to the cloud is an important setup. There should be a proper mitigation plan for each of the identified risk areas. These risks can be technical, non-technical, operational, etc.
- **Time to market:** Planning and identification of a cloud migration time frame, and mapping it with the ongoing enterprise business deliveries is challenging. This time frame is a significant contributor to decision making by enterprise stakeholders.
- **Monitoring and governance:** Fear of lack of control after cloud

migration can be a natural concern. An enterprise is required to have the required in-house skillset to have full control over cloud migration execution. Even the enterprise data must be kept under complete control even though it may move to a third party CSP.

Identification of the right set of strategies is the first and key step towards cloud enablement for an enterprise. Cloud enablement strategies should be mapped with enterprise business objectives and goals. It is important to measure the overall cloud enablement cost, and prepare the payback period and ROI details in the planning phase.

Cloud transformation

Enterprise cloud enablement or transformation starts with the assessment of the existing system of the enterprise. This assessment is required to be done at various levels, as listed below.

- **Platform/Technology:** Assessment of all duplicate/missing/legacy applications, tools or frameworks in the system. Assessment for repetitive manual work, systems or sub-systems to retire, etc.
- **Data:** Assessment of data awareness gaps, data availability gaps, data analysis, etc. This is an important aspect to bring in the right value at the right time associated with the data in the system.
- **People and processes:** Assessment of training requirements, management gaps, financial gaps, facilities gaps, process inefficiencies, information gaps, measurement gaps, compliance needs, security needs, etc.

The assessment of the different components of the enterprise will help to identify the existing cloud maturity level. Broadly, this can be categorised in three stages, as listed below.

- **Initial stage:** This involves enterprise planning for cloud enablement and